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How eddy covariance flux measurements have contributed to our understanding of *Global Change Biology*

Dennis D. Baldocchi

University of California, Berkeley, Berkeley, California

Correspondence
Dennis D. Baldocchi, University of California, Berkeley, Berkeley, CA 94720.
Email: baldocchi@berkeley.edu

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Abstract

A global network of long-term carbon and water flux measurements has existed since the late 1990s. With its representative sampling of the terrestrial biosphere's climate and ecological spaces, this network is providing background information and direct measurements on how ecosystem metabolism responds to environmental and biological forcings and how they may be changing in a warmer world with more carbon dioxide. In this review, I explore how carbon and water fluxes of the world's ecosystem are responding to a suite of covarying environmental factors, like sunlight, temperature, soil moisture, and carbon dioxide. I also report on how coupled carbon and water fluxes are modulated by biological and ecological factors such as phenology and a suite of structural and functional properties. And, I investigate whether long-term trends in carbon and water fluxes are emerging in various ecological and climate spaces and the degree to which they may be driven by physical and biological forcings. As a growing number of time series extend up to 20 years in duration, we are at the verge of capturing ecosystem scale trends in the breathing of a changing biosphere. Consequently, flux measurements need to continue to report on future conditions and responses and assess the efficacy of natural climate solutions.

KEYWORDS

eddy covariance, carbon dioxide, water vapor, evaporation, ecosystem, photosynthesis, ecosystem respiration

1 | INTRODUCTION

The masthead of *Global Change Biology* states that this journal "exists to promote new understanding of the interface between biological systems and all aspects of environmental change that affects a substantial part of the globe." It defines global change as "any consistent trend in the environment—past, present or projected—that affects a substantial part of the globe." In this paper, I discuss how a set of long-term eddy covariance measurements of carbon dioxide and water vapor fluxes has contributed to the study of "global change biology."

One may first ask: why a relatively large number of eddy covariance flux papers have been published in *Global Change Biology*? The answer to this question revolves around such factors as serendipity, opportunity, a paradigm shift in how we measure ecosystem metabolism, and vision of the editors, Chris Field and Stephen Long. These factors converged and resulted in our first set of eddy covariance papers being published in the second volume of this journal. This set of papers was the outcome of the 1995 La Thuile, Italy workshop on "Strategies for Monitoring and Modeling CO_2 and Water Vapor Fluxes over Terrestrial Ecosystems" (Baldocchi, Valentini, Running, Oechel, & Dahlman, 1996). One aim of the workshop was to survey the state of the science and assess our ability to make defensible and reliable long-term eddy covariance measurements of carbon dioxide and water vapor fluxes (Leuning & Judd, 1996; Moncrieff, Malhi, & Leuning, 1996). The other aim was to report on the first set of

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year-long, carbon dioxide and water vapor flux measurements over terrestrial ecosystems (Black et al., 1996; Goulden, Munger, Fan, Daube, & Wofsy, 1996; Greco & Baldocchi, 1996; Valentini et al., 1996). Our hope was to produce long-term records of carbon dioxide and water vapor fluxes, to complement and help interpret the dynamics of the Mauna Loa CO_2 concentration record (Keeling, 1998). We also wanted to produce flux data to parameterize and evaluate coupled carbon–ecosystem–climate models of that era (Cramer et al., 1999; Melillo et al., 1993).

The workshop served as a springboard to establish regional and global networks of long-term flux measurement sites to study the metabolism of the terrestrial biosphere. In the aftermath of the workshop, circa 1997, the Department of Energy funded the Ameriflux network in an ad hoc manner (Novick et al., 2018). Simultaneously, the European Union launched the EUROFLUX network (Aubinet et al., 2000; Valentini et al., 2000), which later morphed into CARBOEUROPE (Dolman et al., 2006; Schulze et al., 2010) and ICOS (Rebmann et al., 2018). Moreover, the global network, FLUXNET, was funded by NASA to provide validation data for the new platform and sensor, EOS/MODIS (Baldocchi et al., 2001; Running et al., 1999). Soon after, our Asian colleagues launched AsiaFlux (Yamamoto et al., 2005) and ChinaFlux (Yu et al., 2006), our Canadian colleagues launched Fluxnet-Canada (Margolis, Flanagan, & Amiro, 2006), OzFlux was initiated in Australia and New Zealand (Beringer et al., 2016), and a confederation of sites was operating in Brazil (Restrepo-Coupe et al., 2013). At this writing, the North American, European, and Asian regional networks have been operating over 20 years and over 900 sites have been registered in FLUXNET (Chu, Baldocchi, John, Wolf, & Reichstein, 2017).

The second question a curious reader may ask is: how do trace gas flux measurements contribute to the study of global change biology?

By integrating long-term, eddy covariance measurements over time and space, we are able to assess ecosystem metabolism on ecosystem time (hours to decades) and space (leaves to communities) scales (Odum, 1969; Vernadsky, 1998). Eddy covariance measurements also produce information on how ecosystem metabolism responds to a plethora of biophysical forcings such as light, temperature, rain, soil moisture, CO_2, phenology, and structural and functional plant traits (Law et al., 2002; Reichstein, Bahn, Mahecha, Kattge, & Baldocchi, 2014; Reichstein et al., 2007; van Dijk, Dolman, & Schulze, 2005). When such measurements are sustained over decades, they can detect how the metabolism of whole ecosystems is responding to secular trends in the environment, like rising CO_2 (Keenan et al., 2013), warming temperatures (Keenan et al., 2014), variable precipitation and soil moisture (Stocker et al., 2018), nitrogen deposition (Fernández-Martínez et al., 2014; Magnani et al., 2007), phenology (Balzarolo et al., 2016; Keenan et al., 2014), and differential land use or land use change (Chen, Dirmeyer, Guo, & Schultz, 2018; Luyssaert et al., 2007; Thornton et al., 2002).

Long-term flux measurements can also capture rare events that can be episodic or extreme (Frank et al., 2015). Episodic and extreme events include insect infestation (Clark, Skowronski, Gallagher, Renninger, & Schäfer, 2012), disease, growing season frost and freeze (Gu et al., 2008), hurricanes and wind throw (Barr, Engel, Smith, & Fuentes, 2012; Knohl et al., 2002), heat spells and drought (Ciais et al., 2005; Schwalm et al., 2012; Wolf et al., 2016), fire (Beringer, Hutley, Tapper, & Cernusak, 2007), flooding and tides (Knox, Windham-Myers, Anderson, Sturtevant, & Bergamaschi, 2018), and logging (Law, Sun, Campbell, Tuyl, & Thornton, 2003; Paul-Limoges, Black, Christen, Nesic, & Jassal, 2015).

The impact of episodic and extreme events on carbon and water fluxes is related to their duration and intensity. They can also exercise direct, indirect, and lagged effects. So far, we have learned that extreme events can have a disproportionate effect on the amount of annual carbon exchange of an ecosystem. One study found that 78% of interannual variation in gross primary productivity (GPP) is induced by extremes that represent 7% of a spatiotemporal domain (Zscheischler et al., 2014).

When flux measurements are paired with a control and either a set of treatments or climate/ecological gradients, they can be used in an experimental manner to study spatial varying factors such as phenology, climate, soil carbon pools, plant functional and structural properties, land use, and disturbance (Anderson-Teixeira, Delong, Fox, Brese, & Litvak, 2011; Curtis & Gough, 2018; Gough, Curtis, Hardiman, Scheuermann, & Bond-Lamberty, 2016; Goulden et al., 2006; Hemes et al., 2019; Reichstein et al., 2014; Richardson et al., 2013; Sanderman, Amundson, & Baldocchi, 2003). Information from networks of flux towers, spanning climate and ecological gradients, can serve as benchmarks for validating and parameterizing coupled carbon–climate models (Bonan, Oleson, Fisher, Lasslop, & Reichstein, 2012; Friend et al., 2007; Williams et al., 2009). Or, flux information can be used with machine learning methods, satellite remote sensing, and gridded climate information to produce spatial maps of net and gross carbon and water fluxes (Jung et al., 2011; Papale & Valentini, 2003; Xiao et al., 2010).

Our ability to make long-term flux measurements is partly due to a co-evolution and the development of solid-state and digital sensors to measure turbulence and trace gas fluctuations (Burba, McDermitt, Anderson, Furtaw, & Eckles, 2010; McDermitt et al., 2010; Wyngaard, 1981), computers and data storage drive, steady funding from many national research organizations, and persistence by legions of scientists, technicians, postdocs, and graduate students. Credit is also due to the many scientists willing to share data and to the teams of data scientists, who have produced data systems that vet the data, provide quality assurance and control, and distribute the data among the wider scientific community (Agarwal et al., 2010; Papale et al., 2012; Pastorello et al., 2017). The sharing and public accessibility of flux data is a new paradigm that has evolved since the 1990s and it has been demonstrated to increase the impact of the eddy covariance flux research (Bond-Lamberty, 2018; Dai et al., 2018). Moreover, data sharing and public distribution have enabled new machine learning methods to proliferate (Beer et al., 2010; Jung et al., 2011; Reichstein et al., 2018; Xiao et al., 2014), giving us the ability to evaluate metabolic fluxes virtually "everywhere, all the time." Consequently, the products from the network of flux sites are proving to be stronger than the collective sum of individual and independent site studies. My only lament is that the scope of the dataset could be broader if more teams shared data.

This capacity, to integrate carbon fluxes in time and space from a sparse network, was not expected when the flux networks were formed in the 1990s. Then, machine learning was in its infancy, and was in the domain of computer scientists (Chhabra, 1993; Schalkoff, 1997). Today, we have a zoo of machine learning methods (artificial neural networks, regression trees) of various degrees of complexity and depth at our disposal (Reichstein et al., 2019). Nor was high resolution, spatial/temporal meteorological available to upscale functional relations between carbon fluxes and environmental variables; the MODIS sensor, on the EOS/TERRA/AQUA satellites, had not been launched (Running et al., 1999). Moreover, gridded climate datasets, for example, ISLSCP (Sellers et al., 1996), Daymet (Thornton, Running, & White, 1997), PRISM (Daly et al., 2000), and CRU, (New, Hulme, & Jones, 1999) were in their infancy.

The intent of this article is to provide an overview of the lessons learned from long-term carbon and water flux measurements, with regard to global change biology. In this review, I will report on general findings with regard to how carbon and water fluxes of ecosystem respond to a suite of covarying environmental factors, like sunlight, temperature, rainfall, soil moisture, and carbon dioxide. Using data from cross-site syntheses, I will discuss how fluxes may respond to biological factors such as phenology, structural and functional traits, disturbance, and land management. I also will distill results on annual sums and long-term trends in carbon and water fluxes from various ecological and climate spaces, and trends in these fluxes.

In forging this discussion, I draw examples from a selection papers written by the members of the FLUXNET community, with many papers published in this journal. This review is not intended to be exhaustive, but rather overarching and concise in scope.

2 | WHAT HAVE WE LEARNED ABOUT CARBON FLUXES?

2.1 | Environmental responses

The environmental physiological literature is rich with studies on how gas exchange of individual leaves responds to environmental factors such as light, temperature, CO_2, vapor pressure deficit, and soil moisture (Berry & Bjorkman, 1980; Larcher, 1975; Pearcy & Ehleringer, 1984). With the advent of eddy covariance systems, one of the first questions asked and answered was how do whole ecosystems, collections of plants and leaves, respond to environmental factors?

To address this question without artifacts, it was pertinent that practitioners develop methods that partitioned net carbon exchange into it components, gross photosynthesis and ecosystem respiration (Falge et al., 2002; Reichstein et al., 2005). In the following, we describe some examples of studies that deconstruct the biophysical controls on ecosystem carbon exchange.

2.1.1 | Light

Leaf photosynthesis is known to saturate at high light levels (Marshall & Biscoe, 1980). Conversely, the earliest flux studies concentrated on managed agricultural ecosystems and reported that productivity, or net ecosystem carbon exchange, was a linear function of absorbed sunlight (Monteith, 1977; Ruimy, Jarvis, Baldocchi, & Saugier, 1995). These early papers were transformative and provided the foundational basis for using light use efficiency models to compute global photosynthesis in that era (Maisongrande, Ruimy, Dedieu, & Saugier, 1995). As data accumulated, we found that the relation between ecosystem photosynthesis and absorbed light is not as simple as first thought. Direct eddy covariance CO_2 measurements of over a variety of native forest and grassland ecosystems found that rates of net carbon exchange saturated with increasing light (Baldocchi & Harley, 1995; Gilmanov et al., 2010). At individual sites, the degree of this nonlinear response varies with leaf area index, soil moisture deficits, and photosynthetic capacity (Baldocchi & Amthor, 2001; Gilmanov et al., 2010; Leuning, Kelliher, Depury, & Schulze, 1995; Xu & Baldocchi, 2004).

We may be on the verge of replacing the inference of canopy photosynthesis by light use efficiency models with an alternative paradigm. A growing set of studies is quantifying how canopy photosynthesis scales linearly with solar-induced fluorescence (Magney et al., 2019; Porcar-Castell et al., 2014; Wood et al., 2017) or with near-infrared radiation reflected by vegetation (Badgley, Field, & Berry, 2017; Zeng et al., 2019). By measuring these variables on towers or satellites, one has alternative ways to upscale photosynthesis in time and space.

Changes in the loading of aerosols in the atmosphere from air pollution and fires have caused global dimming (Liepert, 2002; Stanhill & Cohen, 2001) and brightening (Wild, 2012) of sunlight received at the Earth's surface changes their direct and diffuse fractions (Oliphant & Stoy, 2018). A set of eddy covariance flux studies found that an increase in diffuse light by clouds, humidity, and aerosols increases the light use efficiency of the ecosystem, for example, the slope of the CO_2 flux–light response curve varies with diffuse light (Gu et al., 2002; Hollinger et al., 1994; Niyogi, 2004), supporting theoretical predictions (Jarvis, Miranda, & Muetzelfeldt, 1985; Knohl & Baldocchi, 2008). This increase in light use efficiency compensates for the reduction in global radiation, which otherwise would reduce ecosystem photosynthesis.

The fact that canopy photosynthesis is not a linear function of absorbed light and its sensitivity to light is modified by the fraction of diffuse light may raise concern about using simple linear, light use efficiency models to upscale photosynthesis in time and space, as is being done with satellite remote sensing (Heinsch et al., 2006; Running et al., 1999). Fortunately, complex behavior of photosynthesis-light response functions become linear as one evaluates both the independent and dependent variable in terms of daily integrals (Leuning et al., 1995).

2.1.2 | Temperature

Rising temperature can have a complicated impact on ecosystem metabolism. In cold locales, warming may be viewed as positive, as it increases the rates of enzyme kinetics associated with carbon fixation. In warm and hot locales, additional warming may be viewed as negative. High temperatures can elevate rates of respiration and can cause enzyme denaturization, which inhibits rates of carbon assimilation.

Temperature response of ecosystem carbon exchange is complicated by differential sensitivities of the leaves and the roots and microbes in the soil to temperature. Leaf photosynthesis increases with temperature until a peak temperature (between 20 and 30°C) is reached. Then, it decreases with greater temperatures (Bernacchi, Singsaas, Pimentel, Portis, & Long, 2001; Berry & Bjorkman, 1980; Bjorkman, 1980; Harley & Tenhunen, 1991). More recent data show that the peak temperature of leaf photosynthesis may be plastic and exhibits acclimation (Smith & Dukes, 2017; Way & Yamori, 2014). Soil (roots and microbes) and plant respiration increases exponentially with temperature (Law et al., 2002), as predicted by the Arrhenius function, or respiration can be estimated with a Q_{10} relationship (Davidson, Janssens, & Luo, 2006; Lloyd & Taylor, 1994; Raich & Schlesinger, 1992); the Q_{10} relationship quantifies the relative increase in respiration, like doubling, with a 10° increase in temperature.

What have long-term flux measurements taught us about the response of ecosystem photosynthesis and respiration to temperature? Across ecosystem and climatic gradients, networks of flux studies have shown that the temperature for peak ecosystem photosynthesis is closely related to the mean growing season temperature (Baldocchi et al., 2001; Niu et al., 2012); the key conclusion is that the temperature for peak ecosystem photosynthesis is not static and exhibits acclimation in time and space.

A subset of eddy covariance flux studies have reported high Q_{10} values (>3) for ecosystem respiration (Janssens & Pilegaard, 2003; Van Dijk & Dolman, 2004). These findings may be suspected as many enzyme kinetic processes that constitute life have Q_{10} values near 2. We suspect these high values at ecosystem scales are artifacts resulting from the confounding effects of plotting data across the growing season (to get a wide enough range in temperature) and a suite of factors that may bolster respiration such as leaf area index, photosynthesis, reproduction, phenology, and root exudates (Curiel Yuste, Janssens, Carrara, & Ceulemans, 2004; Deforest et al., 2006).

Cross-site synthesis work that has removed time-sensitive artifacts found that the Q_{10} coefficient (1.4 ± 0.1; Mahecha et al., 2010), or activation energy (Enquist et al., 2003), of ecosystem respiration is very conservative. Instead, it is the reference rate of respiration that changes seasonally (Reichstein et al., 2005; Xu & Baldocchi, 2004); this dynamic can be induced by changes in leaf area index, reproduction stage, and soil moisture deficits.

There is also spatial adjustment in basal rates of ecosystem respiration among ecosystems. Studies report that cold climate ecosystems respire at a greater rate than warm season ecosystems at the same temperature (Enquist et al., 2003), which is consistent with plant studies (Atkin, Bruhn, Hurry, & Tjoelker, 2005).

2.1.3 | Rainfall and soil moisture

Global warming projections indicate that some regions will become wetter, others drier, and yet others with more variability and extremes in the precipitation (Seneviratne et al., 2010). We know from physiological studies that a reduction in soil moisture or an increase in vapor pressure deficit will cause stomata to close and this closure can reduce transpiration and photosynthesis (Katul, Oren, Manzoni, Higgins, & Parlange, 2012; Novick et al., 2016; Schulze, Kelliher, Korner, Lloyd, & Leuning, 1994). Prolonged drought can cause mortality through carbon starvation and embolism (Goulden & Bales, 2019; McDowell et al., 2008).

Rainfall manipulation studies have been used to study how ecosystem metabolism responds to rain (Weltzin et al., 2003). Evaluating how ecosystem metabolism responds to precipitation with rain exclusion or addition studies is challenging due to sample size, timing, and size of treatments (e.g., rain addition or exclusion) and number of replications (Hanson, 2000; Misson et al., 2009). Extracting information from climatic gradients associated with flux networks is an alternative and, I contend, it could be a better way to explore the parameter space between rainfall and ecosystem metabolism.

Concurrent measurements of soil moisture, ground water, evaporation, and rainfall interception provides useful information on how gross and net carbon fluxes vary among ecosystems in different climates. In one of the first FLUXNET synthesis studies, 64% of the variance in annual gross photosynthesis, for a set of deciduous broad-leaved and evergreen needle-leaf forests in temperate and boreal locations, was explained by the product of annual mean air temperature and the summed difference between evaporation and precipitation (Law et al., 2002). In a following study with more data from across the globe, the product of temperature and annual rainfall explained 71% of the variance in GPP (Luyssaert et al., 2007). Interactions indicated that GPP increased with temperature, given enough precipitation (>800 mm/year), while GPP of dry climates (<800 mm/year) did not prosper with warmer temperatures, nor did GPP of cold climates (<5°C) prosper from more rainfall.

Flux measurement studies across networks of semiarid and Mediterranean ecosystems enable us to study how carbon and water fluxes respond in concert to annual rainfall. One synthesis study used 121 site-years of data from 21 sites, whose precipitation ranged between 100 and 1,000 mm/year. They reported that 100 mm change in evaporation caused a 64 g C m^{-2} $year^{-1}$ change in net ecosystem carbon exchange (Biederman et al., 2016).

Studies that relate changes in carbon and water fluxes to changes in mean soil moisture per se may not always be representative of the physiological water deficits that the plants are experiencing. For many circumstances, predawn water potential or root-weighted soil moisture are more representative measures of the moisture status of the soil. In other circumstances, deep-rooted plants can tap deep sources of water (Miller, Chen, Rubin, Ma, & Baldocchi, 2010; Reichstein et al., 2002; Thompson et al., 2011); this is available moisture that may be below the sensors in the field.

With regard to soil respiration, rain on semiarid ecosystems can induce huge pulses of CO_2, known as the Birch effect (Birch, 1958; Jarvis et al., 2007). The size of the pulse depends on the degree of antecedent photodegradation and the pulse size diminishes with time, following succeeding rain events, as the most labile pools of carbon are lost (Huxman et al., 2004; Ma, Baldocchi, Hatala, Detto, & Curiel Yuste, 2012).

There is also evidence that baseline rates of soil respiration, at a set temperature, decreases as soil moisture declines (Baldocchi, Tang, & Xu, 2006; Irvine & Law, 2002; Jassal, Black, Novak, Gaumont-Guay, & Nesic, 2008; Reichstein et al., 2003). Reductions in soil moisture reduce soil respiration two ways. First, it causes photosynthesis to decline, which reduces the exudates it provides heterotrophic microbes in the soil (Baldocchi et al., 2006; Kuzyakov, 2010). Second, it desiccates microbes, which reduce their metabolic activity and growth.

2.2 | Biological responses

Networks of flux towers have been used to examine inter-relations between carbon fluxes and phenology or plant structural and functional properties. We examine some of those findings next.

2.2.1 | Phenology

One of the strongest and widely seen signals of global change biology is a change in phenology, as noted by an increase in the length of the growing season, either by nature of an earlier and/or later growing season (Cleland, Chuine, Menzel, Mooney, & Schwartz, 2007; Menzel et al., 2006; Penuelas & Filella, 2001).

Carbon flux measurements are adept at indicating the dates associated with the start of the growing season, when full leaf cover is achieved, when senescence starts, and when photosynthesis ends (Gu et al., 2009).

Length of growing season can have strong impacts on the cumulative sum of assimilated carbon. Quantification of this effect has been produced by close collaboration between phenocam and flux networks (Keenan et al., 2014; Richardson, Jenkins, et al., 2007; Toomey et al., 2015). Some key lessons include the following. Greenness indices derived from repeat digital images track the photosynthetic phenology of deciduous forests well (Toomey et al., 2015). Deciduous forests are inclined to leaf out and commence photosynthesis when soil temperature exceeds the value of mean annual temperature (Baldocchi et al., 2005). This effect yields an additional 6 g C m^{-2} day^{-1} of the carbon uptake period across north–south gradients of the deciduous forest biome. In comparison, evergreen needle leaf forests experienced 3.4 g C m^{-2} day^{-1} of extended carbon uptake period (Churkina, Schimel, Braswell, & Xiao, 2005).

We offer a word of caution about the effect of a lengthening growing season on net carbon exchange. An earlier onset of photosynthesis can also dry out the soil profile and lead to summer drought. This sequence can reduce photosynthesis later in the growing season and offset the gain by the early onset (Wolf et al., 2016). Results from the Arctic and Boreal regions, where warming is happening fastest, are showing that early season photosynthesis is being offset by increased respiration toward the end of the growing season (Parmentier et al., 2011; Ueyama, Iwata, & Harazono, 2014).

2.2.2 | Functional and structural properties of ecosystems

Information on structural and functional properties of ecosystems, and their constituent plants, is proving to be superior to information based on plant functional groups (Alton, 2011; Rogers, Medlyn, & Dukes, 2014; Van Bodegom et al., 2012). For example, syntheses on the economic spectrum of leaf traits indicate that over 85% of the variation in leaf photosynthesis can be explained by a few functional and structural traits, such as life span, specific leaf weight, and leaf nitrogen (Reich, Walters, & Ellsworth, 1997; Wright et al., 2004). A question pertinent to flux community includes whether useful information about functional and structural properties of ecosystems can be derived from ecosystem scale flux measurements (Reichstein et al., 2014)?

Structural and functional properties that flux scientists are evaluating include potential leaf area index (Baldocchi & Meyers, 1998), leaf life span (Baldocchi et al., 2010; Novick et al., 2015), light use efficiency (the ratio of canopy photosynthesis to absorbed light; Sims et al., 2005; Stocker et al., 2018; Yuan et al., 2007), carbon use efficiency (the ratio between either autotrophic or ecosystem respiration and gross photosynthesis; Baldocchi & Penuelas, 2019; Collalti & Prentice, 2019; Luyssaert et al., 2007), and water use efficiency (WUE) (the ratio between canopy evaporation and photosynthesis; Beer et al., 2009; Medlyn et al., 2017; Zhou, Yu, Huang, & Wang, 2015). Other scientists (Alton, 2017; Dolman, Moors, & Elbers, 2002; Franks et al., 2018; Gilmanov et al., 2010; Groenendijk et al., 2011; Knauer, Werner, & Zaehle, 2015) are evaluating canopy-level, model parameters like maximum photosynthesis of the canopy photosynthesis-light response curve, the carboxylation velocity of the Farquhar–von Caemmerer–Berry photosynthesis model, or the coefficient of coupled stomatal conductance–photosynthesis models (Collatz, Ball, Grivet, & Berry, 1991; Medlyn et al., 2011).

On ecological timescales, the leaf area or density of vegetation of an ecosystem adjusts with mean annual precipitation (Baldocchi & Meyers, 1998; Sankaran et al., 2005; Scheffer, Holmgren, Brovkin, & Claussen, 2005). In principle, evaporation cannot exceed precipitation and photosynthesis scales with evaporation (Joffre & Rambal, 1993). Hence, drier climates will sustain ecosystems with lower leaf area index, who use less water; the converse is true for wetter climates, until light saturation occurs.

Life span properties, like evergreen and deciduous leaves, have fascinated ecologists for decades (Hollinger, 1992; Mooney & Dunn, 1970). We know that longer living leaves have lower nitrogen and lower photosynthetic capacity (Reich et al., 1997). What are the advantages and disadvantages of being evergreen versus deciduous at ecosystem scales?

A meta-analysis of evergreen and deciduous oaks in Mediterranean type climates found that evergreen and deciduous oak woodlands assimilated and respired similar amounts of carbon while using similar amounts of water. These results suggest that evergreen and deciduous woodlands have specific, and similar, ecological costs in Mediterranean climates (Baldocchi et al., 2010). In sum, the two leaf habits experienced similar efficiencies in carbon use (the change in carbon respired per change in carbon assimilated), water use (the change in carbon assimilation per change in water evaporated), and rainfall use (the change in evaporation per change in rainfall).

This functional similarity between different life spans, in the same locale, may not be universal. Another study compared carbon and water fluxes of a colocated deciduous broad-leaved and evergreen conifer forest in North Carolina, United States (Novick et al., 2015). Over 8 years of measurements, the evergreen forest assimilated 104 g C m^{-2} $year^{-1}$ and evaporated 103 mm/year more than the deciduous forest.

Surveys of light use efficiency show a ranking of values among plant functional types (10 mmol/mol, grassland; 16 mmol/mol, temperature deciduous forest; 19 mmol/mol, temperate evergreen needleleaf; 22 mmol/mol, tropical evergreen broadleaved; 27 mmol/mol, crops) when averaged over the growing season (Groenendijk et al., 2011). Nevertheless, modelers are cautioned from using these data to populate models that compute canopy photosynthesis until more general rules have been developed (Groenendijk et al., 2011).

Another study pooled short-term hourly flux data from grasslands, shrublands, savannas, wetlands, and croplands. These authors found a strong relation between light use efficiency (α, mmol/mol) and maximum rates of canopy assimilation (A_{max}, mg-CO_2 m^{-2} s^{-1}; Gilmanov et al., 2010):

$$\alpha = 30A_{max}^{0.55}. \tag{1}$$

If light use efficiency models, like Equation (1), are applied to assess spatial–temporal variations in carbon assimilation, peak values need to decrease with increasing soil moisture or vapor pressure deficits (Running et al., 2004; Stocker et al., 2018).

When carbon use efficiency is defined as the slope of the regression between R_{eco} and GPP from a global database, its value is near 0.82 for undisturbed sites (Baldocchi & Penuelas, 2019). Greater variability in carbon use efficiency is observed when it is evaluated for a variety of forest functional types; when it is defined as the ratio between mean annual R_{eco} and GPP, it ranges between 0.76 and 0.97 (Luyssaert et al., 2007). Together, these data support the conclusion that autotrophic and heterotrophic respiration is a large and relatively conserved fraction of primary productivity (Baldocchi & Penuelas, 2019; Collalti & Prentice, 2019; DeLucia, Drake, Thomas, & Gonzalez-Meler, 2007; Gifford, 1994; Waring, Landsberg, & Williams, 1998).

In principle, WUE of crops is a function of the humidity deficits (Bierhuizen & Slatyer, 1965; Sinclair, Tanner, & Bennett, 1984). Subsequently, members of the flux community used the inherent WUE to adjust site differences for humidity (Beer et al., 2009); inherent WUE is defined as GPP times vapor pressure deficit divided by evaporation. On an annual basis, inherent WUE varied among forests and herbaceous ecosystems by a factor of 3 and it was best correlated (r^2 = .57) with volumetric water content of the soil at field capacity.

An early question posed by the flux community was if WUE varied among plant functional groups or by climate space? One pioneering study showed that WUE of grasslands (3.4 g-CO_2/kg-H_2O), crops (3.1 g-CO_2/kg-H_2O), and deciduous forests (3.2 g-CO_2/kg-H_2O) were similar and were greater than evergreen (2.4 g-CO_2/kg-H_2O) forests (Law et al., 2002); these ratios were derived from the slope of monthly sums. A newer survey of WUE across FLUXNET sites, based on annual sums, find a different ranking: broadleaved deciduous forest (2.66 g-C/kg-H_2O) > broad-leaved evergreen forest (2.57 g-C/kg-H_2O) ~ needleleaf evergreen (2.57 g-C/kg-H_2O) > grassland (2.3 g-C/kg-H_2O) ~ mixed forest (2.23 g-C/kg-H_2O) > crops (1.77 g-C/kg-H_2O; Tang et al., 2014).

Physiologists have long hypothesized that WUE will improve in a higher CO_2 world due to partial stomatal closure and a conservative maintenance of the ratio between internal and atmospheric CO_2 (Farquhar, Oleary, & Berry, 1982; Jones, 1992; Wong, Cowan, & Farquhar, 1979). Two groups of scientists examined how WUE across a set of temperate and boreal forests has changed as CO_2, temperature, and vapor pressure deficits have increased over two decades (Keenan et al., 2013; Knauer et al., 2017). They concluded that WUE increased by about 2% per year.

2.3 | What is the range of annual net carbon fluxes?

The first annual sums of measured net and gross carbon fluxes reach back to 1990 (Wofsy et al., 1993). Prior to 1990, our understanding of the gross and net carbon fluxes of ecosystems on annual timescales was coarse at best. Most information was inferential and was derived from estimates of net primary productivity (Lieth, 1973; Woodwell & Whittaker, 1968); net primary productivity was based on allometric relationships between stem diameter and biomass and measurements of change in stem diameter, litterfall, and soil respiration (Clark et al., 2001). While these approaches could be conducted cheaply and in remote areas, they suffered from sampling and inference errors and the representativeness of their spatial and temporal integration was lacking.

In the intervening 30 years, we have compiled over 2,000 site-years of published sums of annual net and gross ecosystem CO_2 exchange from across the globe. The earliest measurements in this body of published data are providing us with a baseline on the magnitude of net and gross annual carbon exchange of ecosystems spanning the globe's climate and ecological spaces. Contemporary and future flux measurements will enable us to detect if and how fast their carbon fluxes are changing with time. Flux information is also critical information to guide the management and policy using of ecosystems as natural carbon sinks if they are to be used as natural climate solutions (Baldocchi & Penuelas, 2019; Griscom et al., 2017).

2.4 | Are trend's in net carbon fluxes being detected in a changing world?

One question we can ask is: what is the population of annual sums of net ecosystem carbon exchange, measured across the globe, and has it changed on decadal timescales? Figure 1 shows a comparison of the probability density functions for the published values of annual net ecosystem exchange (NEE) before and since 2008; the histograms were fitted with a kernel density function. The expected value of the probability distribution, and its standard deviation, for the first decadal-plus period is −153 ± 281 g C m^{-2} $year^{-1}$. The expected value for the second decadal period is −180 ± 243 g C m^{-2} $year^{-1}$. We applied the nonparametric Kruskal–Wallis ANOVA to compare kernel density fits of the data. We draw the conclusion that the statistics of these two populations of carbon flux measurements did not change significantly over the course of adjacent decades, despite a growing and changing number of sites with different climates, management, and functional types and a warming world with more CO_2 ($p > X^2 = 0.79$). Consequently, Figure 1 suggests that the statistics of carbon fluxes from a global network, with several hundred site-years from a representative selection of climates and ecosystems (Chu et al., 2017), is statistically robust. Therefore, continuing carbon flux measurements from a network, this size, or greater, may be suitable for detecting future change in global metabolism.

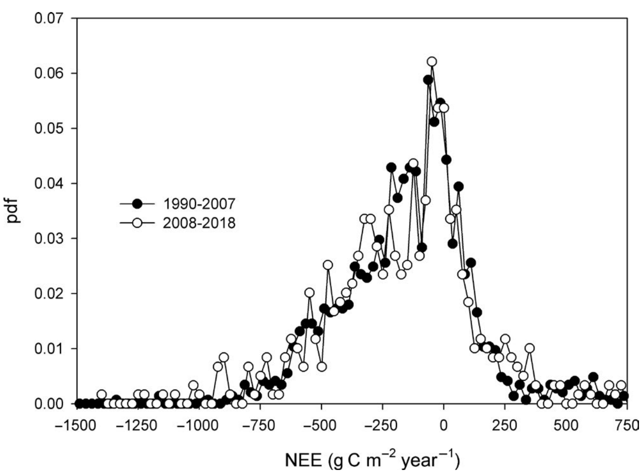

FIGURE 1 Probability density function (pdf) of published data on annual sums of net ecosystem carbon exchange collected before and after 2008. Measurements were fitted with kernel density function. The sample size for data before 2008 was 1,452 site-years, and after 2008, it was 596 site-years. Negative flux densities determine rates of carbon removal from the atmosphere

Based on satellite data starting around the 1980s (Tucker, Fung, Keeling, & Gammon, 1986) and the CO_2 concentration monitoring data, starting in the late 1950s (Keeling, 1998), there is inferential evidence that the metabolism of the biosphere is changing. For example, a set of studies has reported a change in the amplitude of the seasonal variation in atmospheric CO_2 (Ballantyne, Alden, Miller, Tans, & White, 2012; Forkel et al., 2016; Graven et al., 2013). Others are reporting a greening of the biosphere (Park et al., 2016; Tucker et al., 2001). One of the newest papers on this topic conclude that the relative increase in terrestrial global primary productivity (0.08% per year) is less than the trend in greening (0.23% per year; Zhang, Song, Band, & Sun, 2019). Furthermore, this observation of greening is controversial and may be an artifact due to NOAA satellite orbit changes and MODIS sensor degradation (Jiang et al., 2017).

What trends in ecosystem metabolism are being detected from the longest time series of direct carbon flux measurements? To answer this question, researchers must answer three associated questions. The first question relates to how big is the year to year variability of net and gross annual carbon fluxes, relative to the detection limit of a trend? The second question is how sensitive are carbon fluxes are to CO_2, temperature, and in some regions, rainfall or soil moisture and are secular trends in these forcings great enough to induce an observable trend in carbon fluxes? The third overarching question is ecological and addresses whether a detected trend in CO_2 exchange is due to time since disturbance or land use change (Curtis & Gough, 2018; Gough et al., 2016; Odum, 1969)?

2.4.1 | Separating signal from noise

To answer the first question about trend detection, we must first ask: are the published datasets long enough to detect trends given the inter-annual variability and uncertainties in the measurement of annual sums? We conducted a trend analysis, in conjunction with a literature review on interannual variability of carbon fluxes (Baldocchi, Chu, & Reichstein, 2018). We found that temporal trends in net ecosystem carbon exchange must exceed 10 g C m^{-2} $year^{-2}$ to be significantly different than zero if the time series is at least a decade in duration and the measurement uncertainty is plus/minus 60 g C m^{-2} $year^{-1}$.

Figure 2 shows a histogram of time trends from individual sites, for example, slopes, in NEE, determined from published time series that are a decade or longer (Aguilos, Hérault, Burban, Wagner, & Bonal, 2018; Anić et al., 2018; Babel et al., 2017; Clark, Gholz, & Castro, 2004; Dold et al., 2017; Elbers, Jacobs, Kruijt, Jans, & Moors, 2011; Froelich, Croft, Chen, Gonsamo, & Staebler, 2015; Ge, Kellomäki, Zhou, Wang, & Peltola, 2011; Granier, Bréda, Longdoz, Gross, & Ngao, 2008; Grunwald & Bernhofer, 2007; Helfter et al., 2015; Ilvesniemi et al., 2009; Ma, Baldocchi, Wolf, & Verfaillie, 2016; Pilegaard, Ibrom, Courtney, Hummelshøj, & Jensen, 2011; Soloway, Amiro, Dunn, & Wofsy, 2017; Sulman, Roman, Scanlon, Wang, & Novick, 2016; Tamrakar, Rayment, Moyano, Mund, & Knohl, 2018; Urbanski et al., 2007; van Gorsel et al., 2013; Wilkinson, Eaton, Broadmeadow, & Morison, 2012). Most of the trends are clustered between −70 and

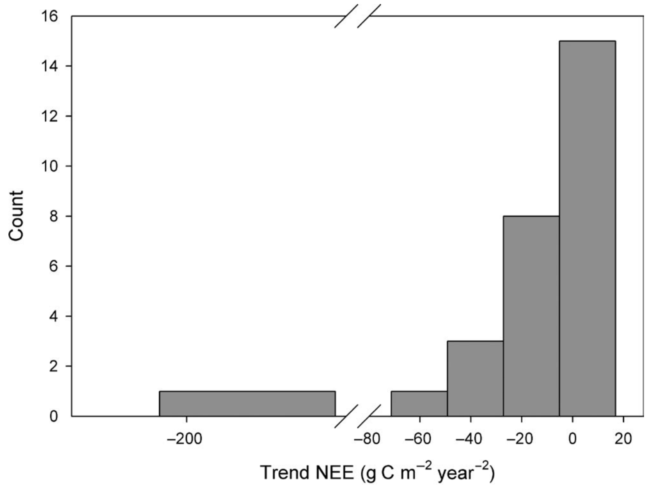

FIGURE 2 Trends in net ecosystem exchange (NEE) derived from published time series a decade in length, or longer. Sources: (Aguilos et al., 2018; Anić et al., 2018; Babel et al., 2017; Clark et al., 2004; Dold et al., 2017; Elbers et al., 2011; Froelich et al., 2015; Ge et al., 2011; Granier et al., 2008; Grunwald & Bernhofer, 2007; Helfter et al., 2015; Ilvesniemi et al., 2009; Ma et al., 2016; Pilegaard et al., 2011; Soloway et al., 2017; Sulman et al., 2016; Tamrakar et al., 2018; Urbanski et al., 2007; van Gorsel et al., 2013; Wilkinson et al., 2012)

20 g C m^{-2} $year^{-2}$. The mean trend from this population of studies is −15.8 ± 15.9 g C m^{-2} $year^{-2}$ (95% CI). However, this statistic is biased by a site with an exceedingly large NEE time trend, more negative than −200 g C m^{-2} $year^{-2}$; this datum is from the succession of a logged, rotational pine plantation in Florida (Clark et al., 2004). If we exclude this highly disturbed plantation site, we calculate a mean trend for this population of 26 studies equal to −8.9 ± 7.5 (95% CI) g C m^{-2} $year^{-2}$. From this limited number of long-term carbon flux studies, we conclude that, on average, this subsample of undisturbed sites is not experiencing trends that are significantly different than the detection limit of the method. We are also learning that trends in carbon fluxes can reverse after published time series are extended. This is the case of more recent carbon flux measurements at Harvard forest (Munger, 1991), when compared to the published time series (Urbanski et al., 2007).

2.4.2 | Variability and trend attribution: Physical versus biological forcings

The answer to the second question, regarding variability and trend attribution, is conditional, complex, and unclear. Many physical and biological forcings are changing simultaneously and they can affect the bidirectional fluxes, photosynthesis and respiration, in distinct and offsetting ways in different climates and ecosystems (Niu et al., 2017; Richardson, Hollinger, Aber, Ollinger, & Braswell, 2007). An emerging observation is that environmental factors provide diminishing power to explain temporal variability in carbon fluxes as the timescale extends beyond a year, compared to functional factors (Richardson, Hollinger, et al., 2007). This finding from a single site has been supported by a study using 481 site-years from 65 AmeriFlux sites. This set of workers found that on average biological effects accounted for 57% of the interannual variability in NEE and the residual (43%) was due to climatic effects (Shao et al., 2015).

The third question involves whether a detected trend is an artifact of ecological succession, following disturbance; Odum conjected that net carbon exchange of an ecosystem is dynamic and reaches an equilibrium between carbon gains and losses as an ecosystem experience various levels of succession over time (Odum, 1969).

Chronosequences, based on regional networks of flux sites, have been used to derive information on how ecosystem CO_2 exchange responds to ecosystem succession, following disturbance (Amiro et al., 2010; Besnard et al., 2018; Goulden et al., 2006). Chronosequences are useful for evaluating how long it takes for a disturbed ecosystem to switch from a carbon source to sink, what is the duration to reach maximum sink potential, will the ecosystem become carbon neutral after a given time, and will these time metrics vary with climate and ecosystem functional group? Relatively slow-growing boreal forests take about 10 years, following disturbance, to become carbon sinks again (Amiro et al., 2010). A test of Odum's theory, using data from 39 temperate forest stands, found little evidence for a decline in net productivity for mid-succession forests, 100–200 years old, and a slight decline for older forests (Curtis & Gough, 2018; Gough et al., 2016). Another study analyzed data from 126 forest sites and fit eddy covariance measurements of annual net ecosystem productivity (NEP) to the function that is dependent on stand age (Besnard et al., 2018). They found that an empirical model describes that the relation between NEP (g C m^{-2} $year^{-1}$) and age (years) shows an asymptotic behavior:

$$\mathrm{NEP} = -324.7 + 587.7\,(1 - \exp(-0.2\,\mathrm{age})). \tag{2}$$

Their empirical model, which considers forest age, climate, soil properties, nitrogen deposition, and management, explained up to 62% of the spatial–temporal variability across a diverse set of forest sites.

3 | WHAT HAVE WE LEARNED ABOUT EVAPORATION?

Evaporation, through transpiration, is the cost of photosynthesis. Net radiation, precipitation, and surface conductance are three overarching variables used to explain differences in evaporation among ecosystems and climate conditions (Budyko, 1974; Jarvis & McNaughton, 1986). We explore these factors next.

3.1 | Energy versus precipitation

From the laws of physics, an ecosystem cannot evaporate more water than is available in a water-limited state. Nor can it evaporate more water than is possible given how the solar radiation load dictates potential evaporation, in an energy-limited state. Budyko (1974) used water balance data and climatology to develop a relation that quantifies annual evaporation based on the limitations imposed by either energy (as under cloudy, wet climates) or water (as under arid and semiarid climates) availability. The functional form of the Budyko relation expresses annual evaporation (E), normalized by annual precipitation (ppt) as a function of the ratio between annual precipitation (ppt) and potential evaporation (E_{pot}) (Budyko, 1974; Sposito, 2017; Williams et al., 2012):

$$\frac{E}{\text{ppt}} = \frac{1}{\left(1+\left(\text{ppt}/E_{pot}\right)^{n}\right)^{1/n}}. \tag{3}$$

The Budyko model was originally derived for large watersheds using multiple years of data, to minimize the effects of leakage and storage (Sposito, 2017). The power law coefficient, n, in Equation (3) was derived from an analysis of FLUXNET data and is 1.48 (Williams et al., 2012).

3.2 | Surface conductance

Information on canopy scale surface conductance (G_c) helps explain limitations imposed on evaporation by ecosystem structure and function. In principle, the surface conductance is a function of leaf area index, soil moisture, leaf nitrogen, photosynthetic capacity, and pathway (Baldocchi & Meyers, 1998; Kelliher, Leuning, Raupach, & Schulze, 1995; Leuning et al., 1995). In practice, the surface conductance can be assessed by inverting the Penman–Monteith equation (Monteith, 1981). Figure 3 shows a comparison between eddy covariance measurements of latent heat exchange, normalized by the equilibrium rate (Raupach, 2001), and surface conductance. The data represent a range of plant functional types, leaf area index, soil moisture, and leaf nitrogen. As surface conductance increases, the ratio between latent heat exchange and the equilibrium rate ($\lambda E/\lambda E_{eq}$) increases, and reaches a limit near 1.26, as defined by the Priestley–Taylor equation (McNaughton & Spriggs, 1989; Priestley & Taylor, 1972).

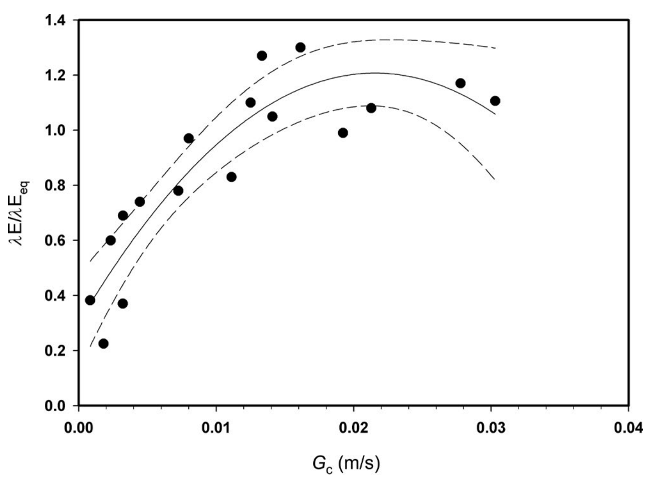

FIGURE 3 Comparison between latent heat exchange (λE), normalized by equilibrium rates (λE_{eq}), and surface conductance (G_c). The surface conductances were computed by inverting the Penman–Monteith equation. This figure was adapted from Valentini, Baldocchi, and Tenhunen (1999), which included data from corn, wheat, potatoes, rice, deciduous broadleaved forests, boreal conifer forests, tropical broadleaved forests, and grasslands. Additional information include results from semiarid, deciduous, broadleaved savanna (Baldocchi & Xu, 2007), annual grassland (Baldocchi et al., 2004), and tule wetlands (Eichelmann et al., 2018)

There remains interest about exploiting linkages between canopy photosynthesis, canopy surface conductance, and canopy stomatal conductance (G_s; Baldocchi & Meyers, 1998; Finnigan & Raupach, 1987; Kelliher et al., 1995). The intent is to develop an independent way to assess G_c for the Penman–Monteith equation. There is hope that one may be able to assess canopy photosynthesis using a light use efficiency model and remote sensing data, then use this information to develop a spatial fields of surface conductance and evaporation.

At the time of this writing, only a few studies have investigated the relation between G_c and the group including gross primary production, relative humidity (rh), and CO_2 concentration (GPP rh/[CO_2]; Baldocchi, Xu, & Kiang, 2004; Dolman et al., 2002). Preliminary results are encouraging and show a good fit. We remind the reader that one has to be careful substituting G_s for the surface conductance (G_c) due to the roles of soil evaporation and leaf area index (Baldocchi & Meyers, 1998; Finnigan & Raupach, 1987; Kelliher et al., 1995). How to best to proceed? Efforts are being focused on how to partition transpiration from evaporation (Stoy et al., 2019) and use this information to estimate the canopy stomatal conductance. There is also need to distinguish when intercepted rainfall is evaporating, as flux measurements are uncertain during such conditions (van Dijk et al., 2015).

3.3 | What is the range in annual evaporation fluxes?

Sites measuring carbon fluxes also measure evaporative fluxes, yet these results of get short shrift and are not reported as often. Independent and direct measurements of stand evaporation are needed to complement and constrain prior assessments of evaporative water use, which have tended to estimate evaporation as a residual of the water budget (Ukkola & Prentice, 2013; Zhang et al., 2011).

Figure 4 shows the histogram of 577 site-years of published data on annual evaporation from ecosystems. The mean and its standard

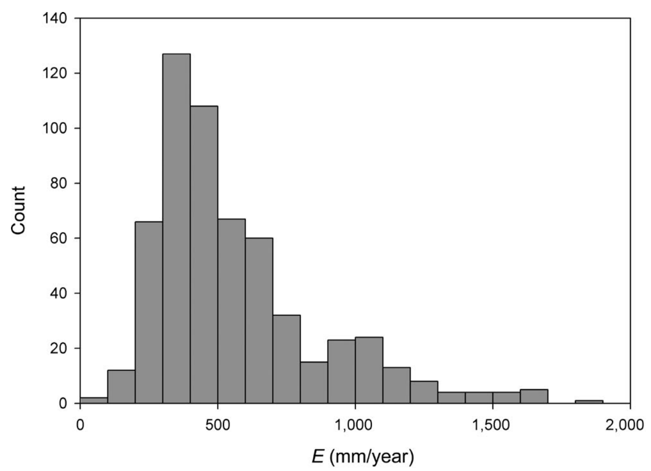

FIGURE 4 Histogram of published values of annual evaporation measured with eddy covariance

deviation are 566 ± 319 mm/year. The histogram is positively skewed with a few sites evaporating more than 1,500 mm/year, which is close to annual evaporation derived from the wettest catchment sites (Zhang, Dawes, & Walker, 2001).

The longest measured and published record of annual evaporation (17 years) is from an evergreen needleleaf forest in Germany. It is showing an upward trend (+16.6 mm/year^2; Babel et al., 2017). A decade long study of evaporation from an evergreen needleleaf forest in Finland (Ilvesniemi et al., 2010) shows a positive trend, too, but with a weaker slope (7.8 mm/year^2) and a 9 year record from another forest in Germany (Grunwald & Bernhofer, 2007) reports a negative slope (−9.35 mm/year^2). All other published records of evaporation in our database are less than 9 years in duration. For perspective, a 32 year water budget study (1972–2004) in Wales showed that the relative difference in evaporation between a forest and grassland decreased with time, as forest evaporation declined with age (−5.6 mm/year^2; Marc & Robinson, 2007).

Whether ecosystems experience more or less evaporation in a warmer world with more CO_2 depends on perspective and scale (Jarvis & McNaughton, 1986; Monteith, 1981). From a thermodynamic and physiological perspective, evaporation from a leaves and vegetated canopies are expected to increase with warming as it increases the saturation vapor pressure of the surface, if stomata remain open. But stomata tend to close with greater CO_2 and humidity deficits (Collatz et al., 1991; Farquhar, 1978), which will mute this increase in transpiration. From a meteorological perspective, warming may decrease evaporation. A warmer surface emits more longwave radiation to the atmosphere, which reduces available energy. In addition, a warmer surface enables more sensible heat to be transferred to the atmosphere, leaving less energy to drive latent heat exchange. When these processes are considered together, a set of feedbacks can act to mute or inhibit the expected positive response of water use by vegetation in a warmer world with more CO_2. Hence, it is not a surprise that directly measured trends in evaporation are relatively small. A key conclusion to be drawn from this time series analysis on evaporation is that decadal- and vicennial-long flux measurements need to be published, to provide a benchmark with the multiple model inferences that are arising.

4 | CONCLUSIONS

Long-term eddy covariance measurements have radically altered how we study and measure the breathing of the terrestrial biosphere in a changing world. The earliest measurements are providing a baseline on how the biosphere functioned in the 1990s. They are also providing direct measurements to parameterize and evaluate a suite of land surface models that are being used to study the carbon cycle and climate (Bonan et al., 2011; Friend et al., 2007; Mercado et al., 2009). And shared data are providing a paradigm shift for upscaling fluxes using a new generation of machine learning methods (Jung et al., 2011).

What is next? If we are to employ natural carbon solutions (Griscom et al., 2017) and monetize them in carbon markets, we need to deploy more and cheap flux systems to monitor their performance. This may require us to deploy low-powered, flux stations in remote and underrepresented regions and transmit data using new technologies. To assess "fluxes everywhere and all the time," there must be growth in handshaking between flux towers and the new generation of hyperspectral and high-resolution satellites, like CubeSat (Aragon, Houborg, Tu, Fisher, & McCabe, 2018), and interpreting these data with deep learning methods (Reichstein et al., 2019) across representative flux footprints. We also need to grow the flux networks to include a suite of greenhouse gases like methane (Knox et al., 2019) and nitrous oxide (Taki, Wagner-Riddle, Parkin, Gordon, & VanderZaag, 2018) and their stable isotopes (Bowling, Sargent, Tanner, & Ehleringer, 2003; Oikawa et al., 2017).

How long should carbon and water flux studies continue? With many sites accumulating time series that are reaching 20 years in duration, their value is virtually priceless, as we cannot turn back the clock. I would, therefore, advocate that we continue to fund and conduct these flux measurements indefinitely. One can argue that much novel and significant science about global change biology remains.

First, we are on the verge of quantifying how the metabolism of the biosphere is changing in a warmer world with more CO_2. Second, extended time series will better enable us to address questions about secular in trends and attributions on metabolic fluxes with more statistical confidence. Conceding that we are living in a changing world, we need to know how whole ecosystems are responding to multifaceted changes in CO_2, temperature, water, and management. Third, many of us who have been working over 20 years continue to find surprises and discoveries by serendipity, which has its own intrinsic merit.

In closing, one can argue that these measurements have the potential to act as the proverbial "canary in the mine." So, we need to future flux measurements of ecosystem metabolism to know if the "canary" is remaining healthy, or not.

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ORCID

Dennis D. Baldocchi https://orcid.org/0000-0003-3496-4919

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